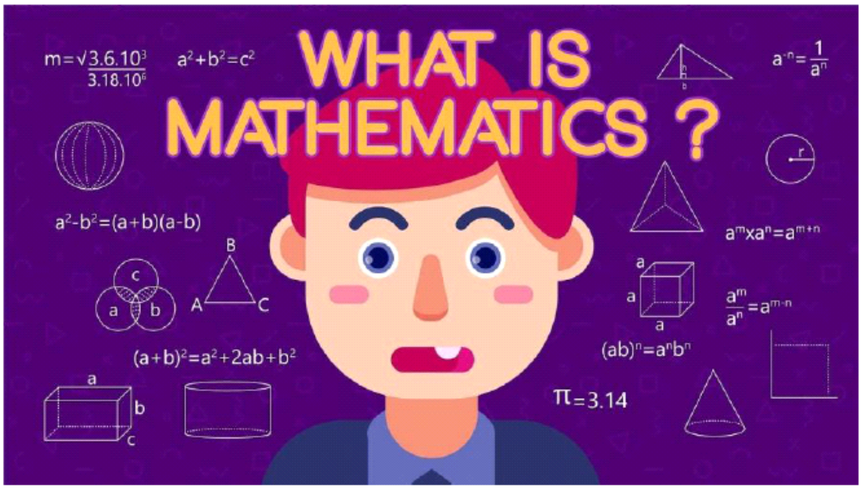
# Lec-01-Week-01

Thursday, 13 August 2026 10:25 am

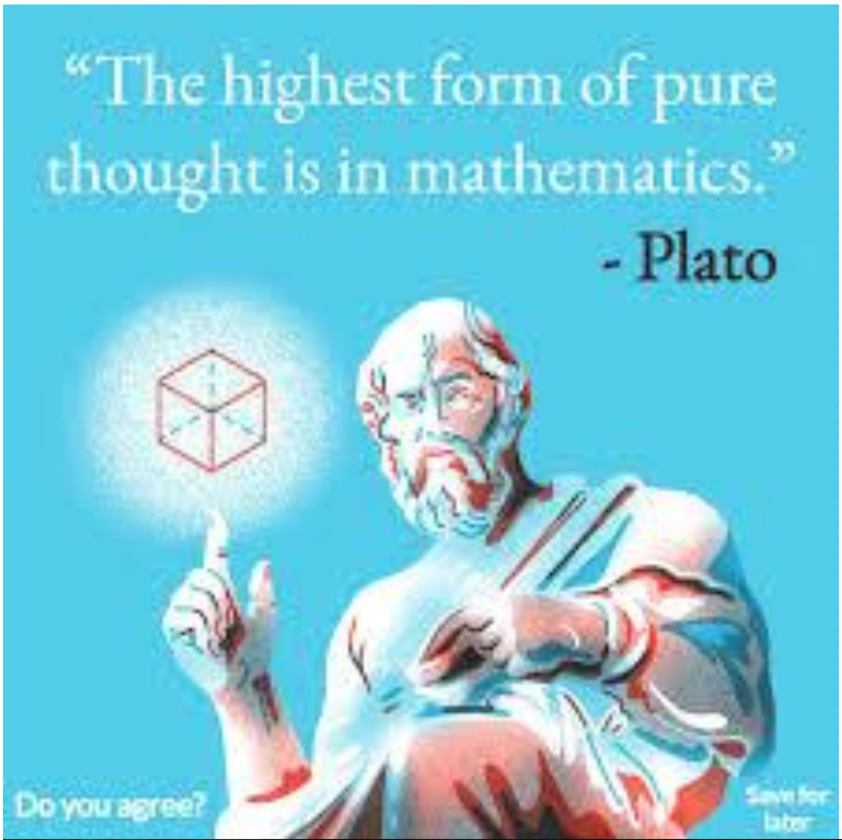
- ## Mathematics



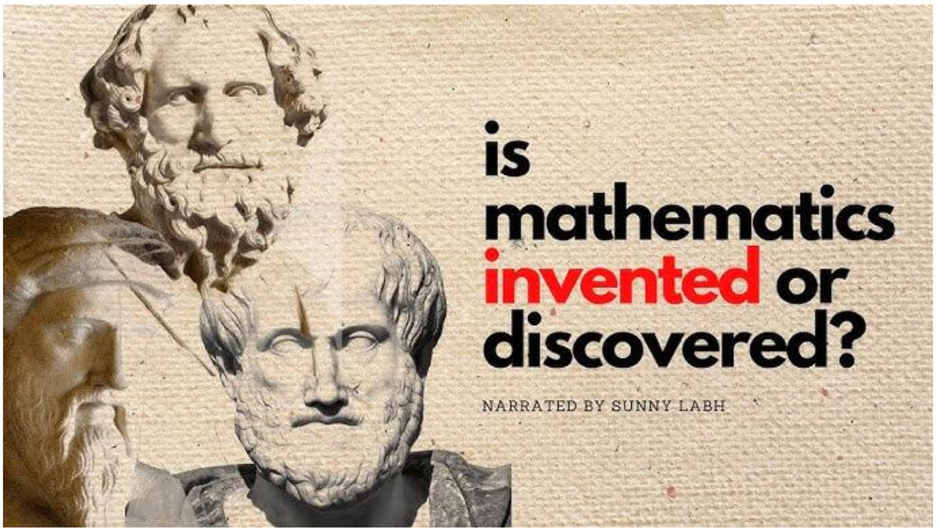

The word *mathematics* comes from [Ancient Greek](#) *máthēma* ([*μάθημα*](#)), meaning "**that which is learnt**", "**what one gets to know**", hence also "study" and "science".

Its [adjective](#) is *mathēmatikós* (μαθηματικός), meaning **"related to learning" or "studious"**, which likewise further came to mean "mathematical".

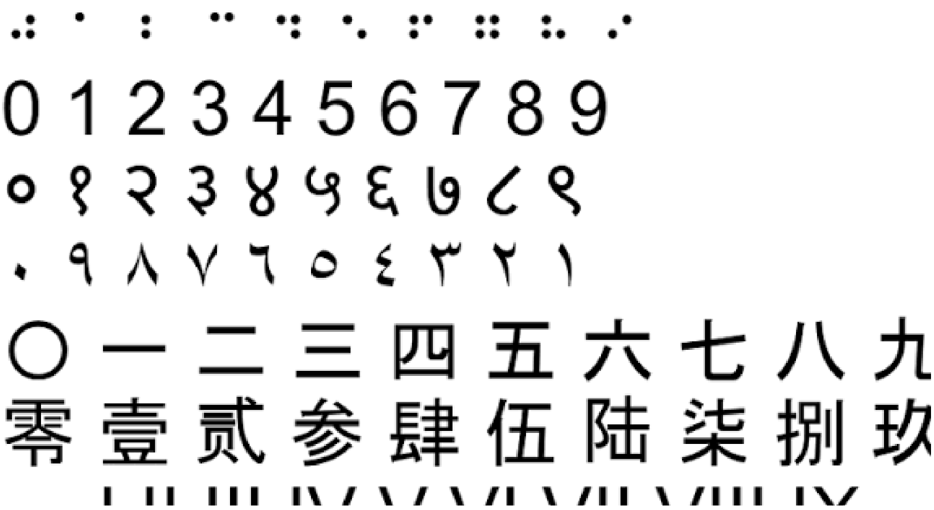
Similarly, one of the two main schools of thought in [Pythagoreanism](#) was known as the *mathēmatikoi* (μαθηματικοί)—which at the time meant "**learners**" rather than "mathematicians" in the modern sense.


"The highest form of pure thought is in mathematics."
- Plato
Do you agree?
Save for later

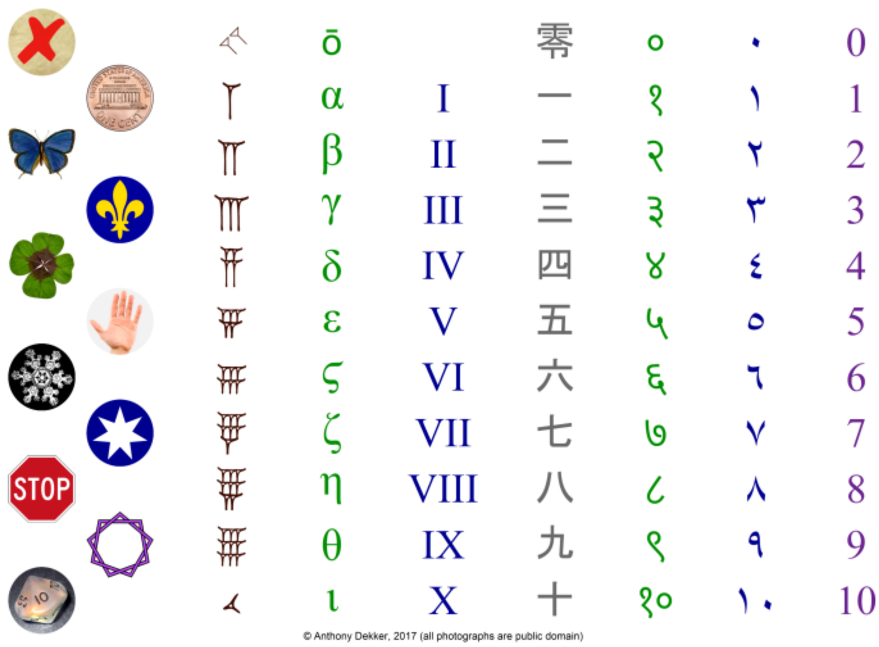

HE WHO IS WITHOUT MATHEMATICS SHALL NOT ENTER
The Caves

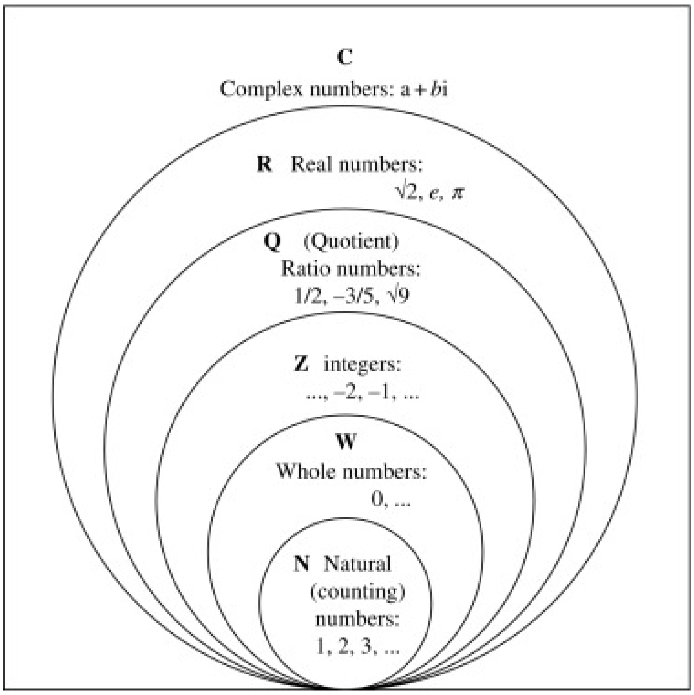


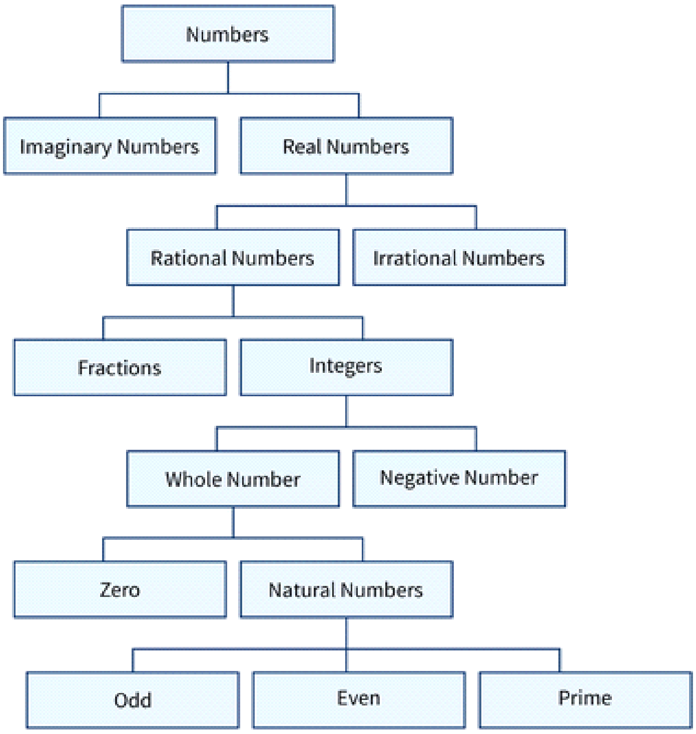

## Numeral system




Babylonian, Greek, Roman, Chinese, Indian, Arabic, and Western numerals






## Numbers, Inequalities, and Absolute Values

Calculus is based on the real number system. We start with the **integers**:

$$\ldots, \quad -3, \quad -2, \quad -1, \quad 0, \quad 1, \quad 2, \quad 3, \quad 4, \quad \ldots$$

Then we construct the **rational numbers**, which are ratios of integers. Thus any rational number $r$ can be expressed as

$$r = \frac{m}{n} \qquad \text{where } m \text{ and } n \text{ are integers and } n \neq 0$$

Examples are

$$\tfrac{1}{2} \qquad -\tfrac{3}{7} \qquad 46 = \tfrac{46}{1} \qquad 0.17 = \tfrac{17}{100}$$

(Recall that division by 0 is always ruled out, so expressions like $\frac{3}{0}$ and $\frac{0}{0}$ are undefined.) Some real numbers, such as $\sqrt{2}$, can't be expressed as a ratio of integers and are therefore called **irrational numbers**. It can be shown, with varying degrees of difficulty, that the following are also irrational numbers:

$$\sqrt{3} \qquad \sqrt{5} \qquad \sqrt[3]{2} \qquad \pi \qquad \sin 1^\circ \qquad \log_{10} 2$$

The set of all real numbers is usually denoted by the symbol $\mathbb{R}$. When we use the word *number* without qualification, we mean "real number."

Every number has a decimal representation. If the number is rational, then the corresponding decimal is repeating. For example,

$$\tfrac{1}{2} = 0.5000\ldots = 0.5\overline{0} \qquad \tfrac{2}{3} = 0.66666\ldots = 0.\overline{6}$$

$$\tfrac{157}{495} = 0.317171717\ldots = 0.3\overline{17} \qquad \tfrac{9}{7} = 1.285714285714\ldots = 1.\overline{285714}$$

(The bar indicates that the sequence of digits repeats forever.) On the other hand, if the number is irrational, the decimal is nonrepeating:

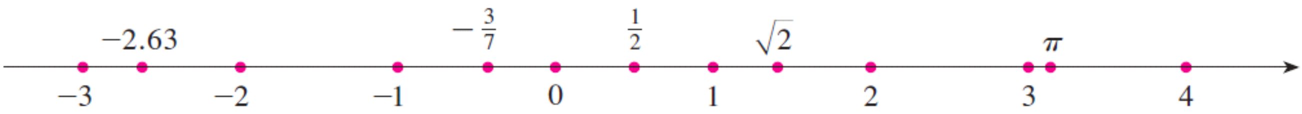
$$\sqrt{2} = 1.414213562373095\ldots \qquad \pi = 3.141592653589793\ldots$$

If we stop the decimal expansion of any number at a certain place, we get an approximation to the number. For instance, we can write

$$\pi \approx 3.14159265$$

where the symbol $\approx$ is read "is approximately equal to." The more decimal places we retain, the better the approximation we get.

**line**, or a **real number line**, or simply a **real line**. Often we identify the point with its coordinate and think of a number as being a point on the real line.




The real numbers are ordered. We say *$a$ is less than $b$* and write $a < b$ if $b - a$ is a positive number. Geometrically this means that $a$ lies to the left of $b$ on the number line. (Equivalently, we say *$b$ is greater than $a$* and write $b > a$.) The symbol $a \leqslant b$ (or $b \geqslant a$) means that either $a < b$ or $a = b$ and is read "$a$ is less than or equal to $b$." For instance, the following are true inequalities:

$$7 < 7.4 < 7.5 \qquad -3 > -\pi \qquad \sqrt{2} < 2 \qquad \sqrt{2} \leqslant 2 \qquad 2 \leqslant 2$$

In what follows we need to use *set notation*. A **set** is a collection of objects, and these objects are called the **elements** of the set. If $S$ is a set, the notation $a \in S$ means that $a$ is an element of $S$, and $a \notin S$ means that $a$ is not an element of $S$. For example, if $Z$ represents the set of integers, then $-3 \in Z$ but $\pi \notin Z$. If $S$ and $T$ are sets, then their **union** $S \cup T$ is the set consisting of all elements that are in $S$ or $T$ (or in both $S$ and $T$). The **intersection** of $S$ and $T$ is the set $S \cap T$ consisting of all elements that are in both $S$ and $T$. In other words, $S \cap T$ is the common part of $S$ and $T$. The empty set, denoted by $\varnothing$, is the set that contains no element.

Some sets can be described by listing their elements between braces. For instance, the set $A$ consisting of all positive integers less than 7 can be written as

$$A = \{1, 2, 3, 4, 5, 6\}$$

We could also write $A$ in *set-builder notation* as

$$A = \{x \mid x \text{ is an integer and } 0 < x < 7\}$$

which is read "$A$ is the set of $x$ such that $x$ is an integer and $0 < x < 7$."

## Intervals

Certain sets of real numbers, called **intervals**, occur frequently in calculus and correspond geometrically to line segments. For example, if $a < b$, the **open interval** from $a$ to $b$ consists of all numbers between $a$ and $b$ and is denoted by the symbol $(a, b)$. Using set-builder notation, we can write

$$(a, b) = \{x \mid a < x < b\}$$

Notice that the endpoints of the interval—namely, $a$ and $b$—are excluded. This is indicated by the round brackets ( ) and by the open dots in Figure 2. The **closed interval** from $a$ to

Notice that the endpoints of the interval—namely, $a$ and $b$—are excluded. This is indicated by the round brackets ( ) and by the open dots in Figure 2. The **closed interval** from $a$ to $b$ is the set

$$[a, b] = \{x \mid a \leqslant x \leqslant b\}$$

Here the endpoints of the interval are included. This is indicated by the square brackets [ ] and by the solid dots in Figure 3. It is also possible to include only one endpoint in an interval, as shown in Table 1.

| Notation | Set description | Picture |
|---|---|---|
| $(a, b)$ | $\{x \mid a < x < b\}$ |   |
| $[a, b]$ | $\{x \mid a \leqslant x \leqslant b\}$ |   |
| $[a, b)$ | $\{x \mid a \leqslant x < b\}$ |   |
| $(a, b]$ | $\{x \mid a < x \leqslant b\}$ |   |
| $(a, \infty)$ | $\{x \mid x > a\}$ |   |
| $[a, \infty)$ | $\{x \mid x \geqslant a\}$ |   |
| $(-\infty, b)$ | $\{x \mid x < b\}$ |   |
| $(-\infty, b]$ | $\{x \mid x \leqslant b\}$ |   |
| $(-\infty, \infty)$ | $\mathbb{R}$ (set of all real numbers) |  |

We also need to consider infinite intervals such as

$$(a, \infty) = \{x \mid x > a\}$$

## Inequalities

When working with inequalities, note the following rules.

**2 Rules for Inequalities**

1. If $a < b$, then $a + c < b + c$.
2. If $a < b$ and $c < d$, then $a + c < b + d$.
3. If $a < b$ and $c > 0$, then $ac < bc$.
4. If $a < b$ and $c < 0$, then $ac > bc$.
5. If $0 < a < b$, then $1/a > 1/b$.

**EXAMPLE 1** Solve the inequality $1 + x < 7x + 5$.

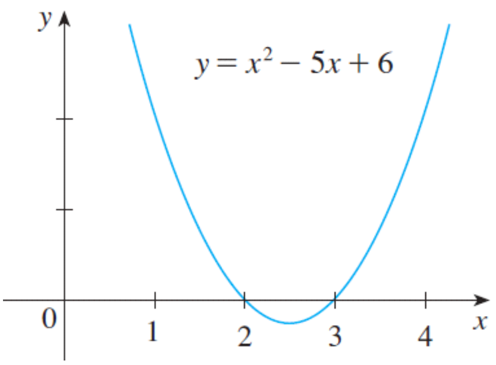
**EXAMPLE 2** Solve the inequalities $4 \leqslant 3x - 2 < 13$.

**EXAMPLE 3** Solve the inequality $x^2 - 5x + 6 \leqslant 0$.

$$(x - 2)(x - 3) \leqslant 0$$

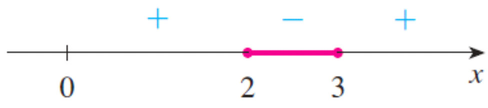
$$(-\infty, 2) \qquad (2, 3) \qquad (3, \infty)$$

$$x \in (-\infty, 2) \quad \Rightarrow \quad x < 2 \quad \Rightarrow \quad x - 2 < 0$$




| Interval | $x - 2$ | $x - 3$ | $(x - 2)(x - 3)$ |
|---|---|---|---|
| $x < 2$ | $-$ | $-$ | $+$ |
| $2 < x < 3$ | $+$ | $-$ | $-$ |
| $x > 3$ | $+$ | $+$ | $+$ |

$$\{x \mid 2 \leqslant x \leqslant 3\} = [2, 3]$$




**EXAMPLE 4** Solve $x^3 + 3x^2 > 4x$.

$$x^3 + 3x^2 - 4x > 0 \qquad \text{or} \qquad x(x-1)(x+4) > 0$$

As in Example 3 we solve the corresponding equation $x(x-1)(x+4) = 0$ and use the solutions $x = -4$, $x = 0$, and $x = 1$ to divide the real line into four intervals $(-\infty, -4)$, $(-4, 0)$, $(0, 1)$, and $(1, \infty)$. On each interval the product keeps a constant sign as shown in the following chart:

| Interval | $x$ | $x-1$ | $x+4$ | $x(x-1)(x+4)$ |
|---|---|---|---|---|
| $x < -4$ | − | − | − | − |
| $-4 < x < 0$ | − | − | + | + |
| $0 < x < 1$ | + | − | + | − |
| $x > 1$ | + | + | + | + |

$$\{x \mid -4 < x < 0 \text{ or } x > 1\} = (-4, 0) \cup (1, \infty)$$


## Absolute Value

The **absolute value** of a number $a$, denoted by $|a|$, is the distance from $a$ to 0 on the real number line. Distances are always positive or 0, so we have

$$|a| \geqslant 0 \qquad \text{for every number } a$$

For example,

$$|3| = 3 \qquad |-3| = 3 \qquad |0| = 0 \qquad |\sqrt{2} - 1| = \sqrt{2} - 1 \qquad |3 - \pi| = \pi - 3$$

$$|a| = a \qquad \text{if } a \geqslant 0$$

$$|a| = -a \qquad \text{if } a < 0$$

**EXAMPLE 5** Express $|3x - 2|$ without using the absolute-value symbol.

SOLUTION

$$|3x-2| = \begin{cases} 3x-2 & \text{if } 3x-2 \geqslant 0 \\ -(3x-2) & \text{if } 3x-2 < 0 \end{cases}$$

$$= \begin{cases} 3x-2 & \text{if } x \geqslant \frac{2}{3} \\ 2-3x & \text{if } x < \frac{2}{3} \end{cases}$$

$$\sqrt{a^2} = |a|$$

$\boxed{5}$ **Properties of Absolute Values** Suppose $a$ and $b$ are any real numbers and $n$ is an integer. Then

**1.** $|ab| = |a||b|$ **2.** $\left|\dfrac{a}{b}\right| = \dfrac{|a|}{|b|} \quad (b \neq 0)$ **3.** $|a^n| = |a|^n$

$\boxed{6}$ Suppose $a > 0$. Then

**4.** $|x| = a$ if and only if $x = \pm a$

**5.** $|x| < a$ if and only if $-a < x < a$

**6.** $|x| > a$ if and only if $x > a$ or $x < -a$

**EXAMPLE 6** Solve $|2x - 5| = 3$.

SOLUTION By Property 4 of $\boxed{6}$, $|2x - 5| = 3$ is equivalent to

$$2x - 5 = 3 \qquad \text{or} \qquad 2x - 5 = -3$$

So $2x = 8$ or $2x = 2$. Thus $x = 4$ or $x = 1$.

**EXAMPLE 7** Solve $|x - 5| < 2$.

SOLUTION 1 By Property 5 of $\boxed{6}$, $|x - 5| < 2$ is equivalent to

$$-2 < x - 5 < 2$$

Therefore, adding 5 to each side, we have

$$3 < x < 7$$

and the solution set is the open interval $(3, 7)$.

**EXAMPLE 8** Solve $|3x + 2| \geqslant 4$.

SOLUTION By Properties 4 and 6 of $\boxed{6}$, $|3x + 2| \geqslant 4$ is equivalent to

$$3x + 2 \geqslant 4 \qquad \text{or} \qquad 3x + 2 \leqslant -4$$

In the first case $3x \geqslant 2$, which gives $x \geqslant \frac{2}{3}$. In the second case $3x \leqslant -6$, which gives $x \leqslant -2$. So the solution set is

$$\left\{x \mid x \leqslant -2 \text{ or } x \geqslant \tfrac{2}{3}\right\} = (-\infty, -2] \cup \left[\tfrac{2}{3}, \infty\right)$$

$\boxed{7}$ **The Triangle Inequality** If $a$ and $b$ are any real numbers, then

$$|a + b| \leqslant |a| + |b|$$

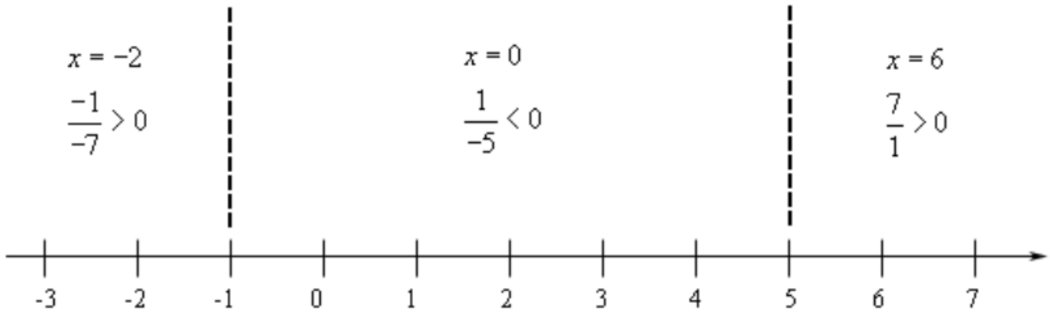
**EXAMPLE 9** If $|x - 4| < 0.1$ and $|y - 7| < 0.2$, use the Triangle Inequality to estimate $|(x + y) - 11|$.

SOLUTION In order to use the given information, we use the Triangle Inequality with $a = x - 4$ and $b = y - 7$:

$$\begin{aligned} |(x + y) - 11| &= |(x - 4) + (y - 7)| \\ &\leqslant |x - 4| + |y - 7| \\ &< 0.1 + 0.2 = 0.3 \end{aligned}$$

Thus $$|(x + y) - 11| < 0.3$$

***Example 1*** Solve $\dfrac{x+1}{x-5} \leq 0$.

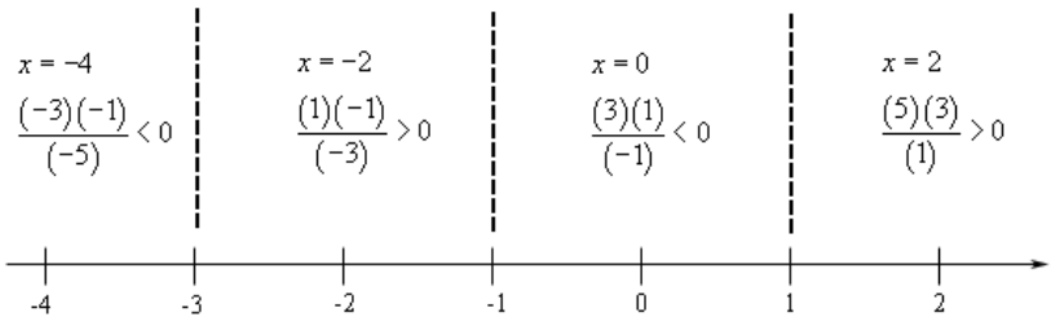



$-1 \leq x < 5 \qquad [-1, 5)$

***Example 2*** Solve $\dfrac{x^2 + 4x + 3}{x - 1} > 0$.

$$\frac{(x+1)(x+3)}{x-1} > 0$$

$$\text{numerator}: x = -1, \quad x = -3 \qquad \text{denominator}: x = 1$$




$-3 < x < -1 \quad \text{and} \quad 1 < x < \infty$

$(-3, -1) \quad \text{and} \quad (1, \infty)$

***Example 3*** Solve $\dfrac{x^2 - 16}{(x-1)^2} < 0$.

***Example 4*** Solve $\dfrac{3x + 1}{x + 4} \geq 1$.

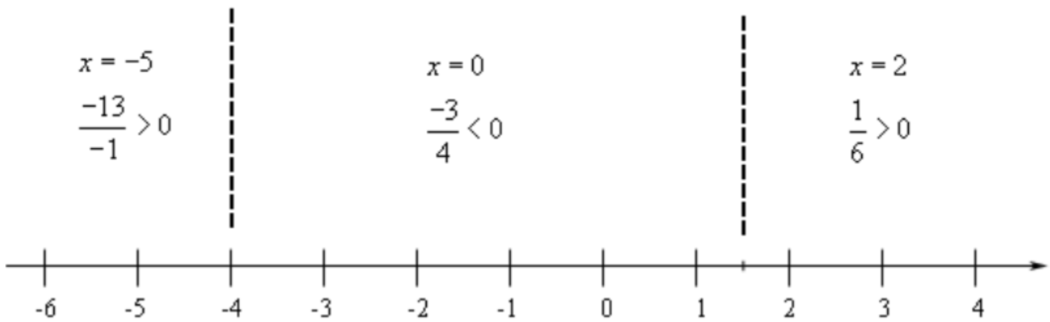
$$\frac{3x+1}{x+4}-1 \geq 0$$

$$\frac{3x+1}{x+4}-\frac{x+4}{x+4} \geq 0$$

$$\frac{3x+1-(x+4)}{x+4} \geq 0$$

$$\frac{2x-3}{x+4} \geq 0$$

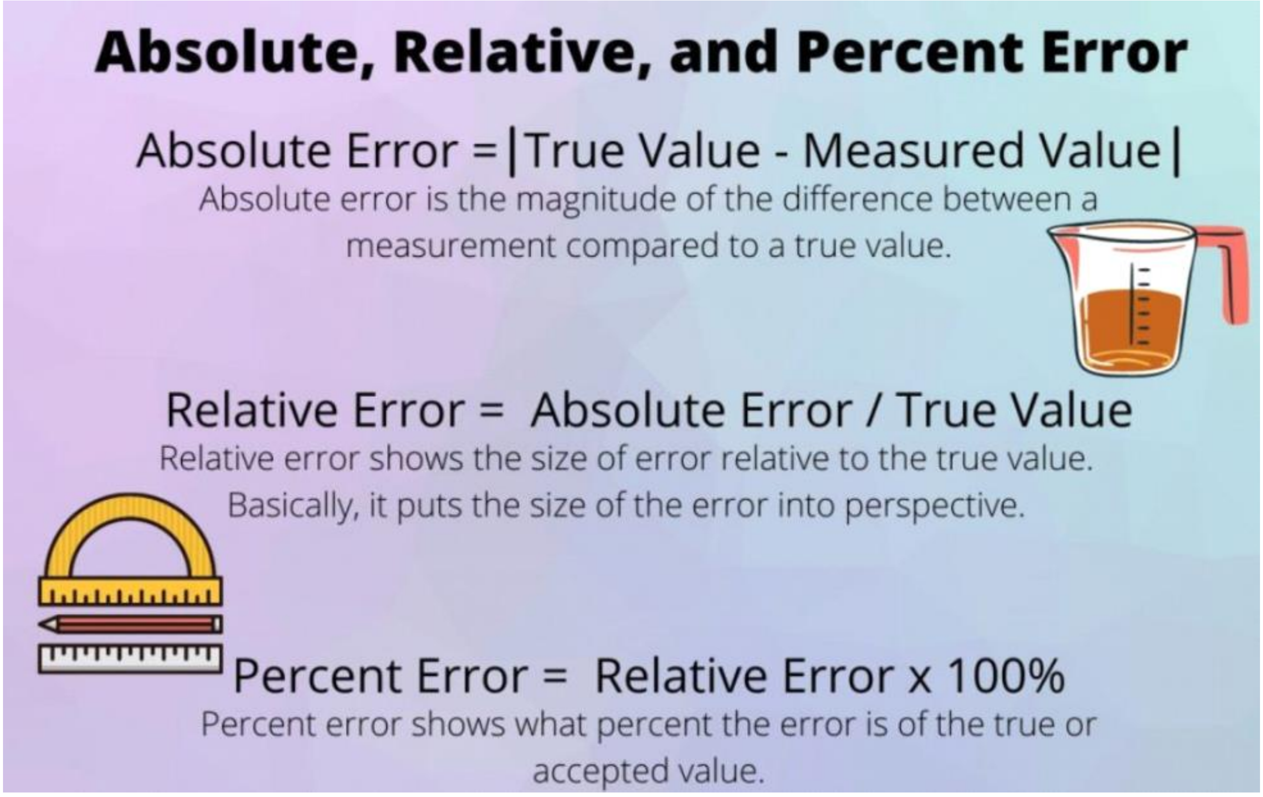
$$\text{numerator} : x = \frac{3}{2} \qquad \text{denominator} : x = -4$$




$$-\infty < x < -4 \quad \text{and} \quad \frac{3}{2} \leq x < \infty$$

$$(-\infty,-4) \quad \text{and} \quad \left[\frac{3}{2},\infty\right)$$

***Example 5*** Solve $\dfrac{x-8}{x} \leq 3-x$.




## Basic Formulas

- **Absolute Error ($E_A$):**
  $E_A = |x_{\text{meas}} - x_{\text{true}}|$[0.6.7]
- **Relative Error ($E_R$):**
  $E_R = \frac{E_A}{|x_{\text{true}}|} = \frac{|x_{\text{meas}} - x_{\text{true}}|}{|x_{\text{true}}|}$ [0.6.7, 0.6.10]
- **Percentage Error ($E_{\%}$):**
  $E_{\%} = E_R \times 100\%$[0.6.7]